



OBJECT ORIENTED SOFTWARE ENGINEERING

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Agile Process Model

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SCRUM - Agile methodology

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SCRUM



- Originally proposed by **Schwaber and Beedle**.
- “**Scrum** is a **framework** for developing complex products and systems. It is grounded in empirical process and control theory. Scrum employs an **iterative and incremental** approach to optimize predictability & control risk”
– Ken Schwaber.

- Scrum—distinguishing features
 - Development work is partitioned into “**packets**”.
 - **Testing and documentation are on-going** as the product is constructed.
 - Work occurs in “**sprints**” and is derived from a “**backlog**” of existing requirements.
 - **Meetings are very short** and sometimes conducted without chairs. A team facilitator, called a **Scrum master**, leads the meeting and assesses the responses from each person.
 - “**demos**” are delivered to the customer with the time-box allocated

- Scrum incorporates a set of process patterns that emphasize
 - **project priorities,**
 - **compartmentalized work units,**
 - **communication, and**
 - **frequent customer feedback.**

Scrum roles

There are 3 Roles :

1. **Product Owner**
2. **Team (development team)**
3. **Scrum Master**

1. Product Owner:

- A Scrum Product Owner is accountable for **maximizing the value of the product** resulting from the work of the Scrum Team.
- How this is done may vary widely across organizations, Scrum Teams, and individuals.
- The Product Owner is a person, but not a committee. The Product Owner may **represent the needs of many stakeholders** in the Product Backlog.
 - Those wanting to change the Product Backlog can do so by trying to convince the Product Owner.

Product Owner...

- The Product Owner is also accountable for effective Product Backlog management, which includes:
 - Developing and explicitly communicating to achieve the Product goal;
 - Creating and clearly communicating Product Backlog items;
 - Ordering Product Backlog items; and,
 - Ensuring that the Product Backlog is transparent, visible and understood.

2. Development team

- The Development Team decides how to accomplish the work set forth by the Product Owner. Development Teams are structured and empowered to organize and manage their own work.

Development teams have the following **characteristics**:

- They are **self-organizing**. No one (not even the ScrumMaster) tells the development team how to turn Product Backlog into Increments of potentially releasable functionality;
- Development teams are **cross-functional**, with all the skills as a team necessary to create a product Increment;

Development team...

- Scrum recognizes **no titles for development team members**, regardless of the work being performed by the person;
- Scrum recognizes **no sub-teams** in the development team, regardless of domains that need to be addressed like testing, architecture, operations, or business analysis;
- Individual development team members may have specialized skills and areas of focus, but accountability belongs to the development team as a whole.

3. Scrum master:

- Scrum master **serves the team** (helping them remove any and all impediments that surface),
- **Protects the team** (from any outside disruption or interference) and
- **Teaches & guides** the team - use of scrum
- He also train both team and product owner about **values and principles** of scrum.

Some of the keywords used in Scrum

- **Product Backlog**—A *prioritized list of project requirements*, the product manager assesses the backlog and updates priorities as required.
- **Sprints**—consist of **work units** that are required to achieve each requirements defined in the backlog that must be fit into a **predefined time-box(typically 30 days)**.
- Changes are not introduced during the sprint. Hence, the sprint allows team members to work in a short-term, but stable environment

- **Daily Scrum Meeting:** The Daily Scrum is the key inspect and adapt meeting during a Sprint.
- During the Sprint execution , the scrum Team meets every day, for **Daily Scrum meeting** and **inspects** the **progress** and ensures communication flow inside the Team. **It is a short (15 minutes) meeting.**
- **Time box:** 15 Minutes
- **Attendees:** Scrum Master, Product Owner(Optional) and Scrum development team

- Daily scrum meeting is held **at the same time at same place each day.**
- During the meeting, each Team member explains:
 - What have I accomplished since the last meeting?
 - What am I going to do before the next meeting?
 - What obstacles are in my way?

Sprint backlog:

- The Sprint Backlog is the set of **Product Backlog items selected for the Sprint**, plus a plan for delivering the product Increment and realizing the Sprint Goal.
- The Sprint Backlog is a forecast by the Team about what functionality will be made available in the next Increment and the work needed to deliver that functionality as a working product Increment

- The Sprint Backlog is a plan with enough detail that can be understood but the Team to track in the Daily Scrum. The **Team modifies the Sprint Backlog** throughout the Sprint.
- Backlog emerges during the Sprint. This emergence occurs as the Team works through the plan and learns more about the work needed to achieve the Sprint Goal.

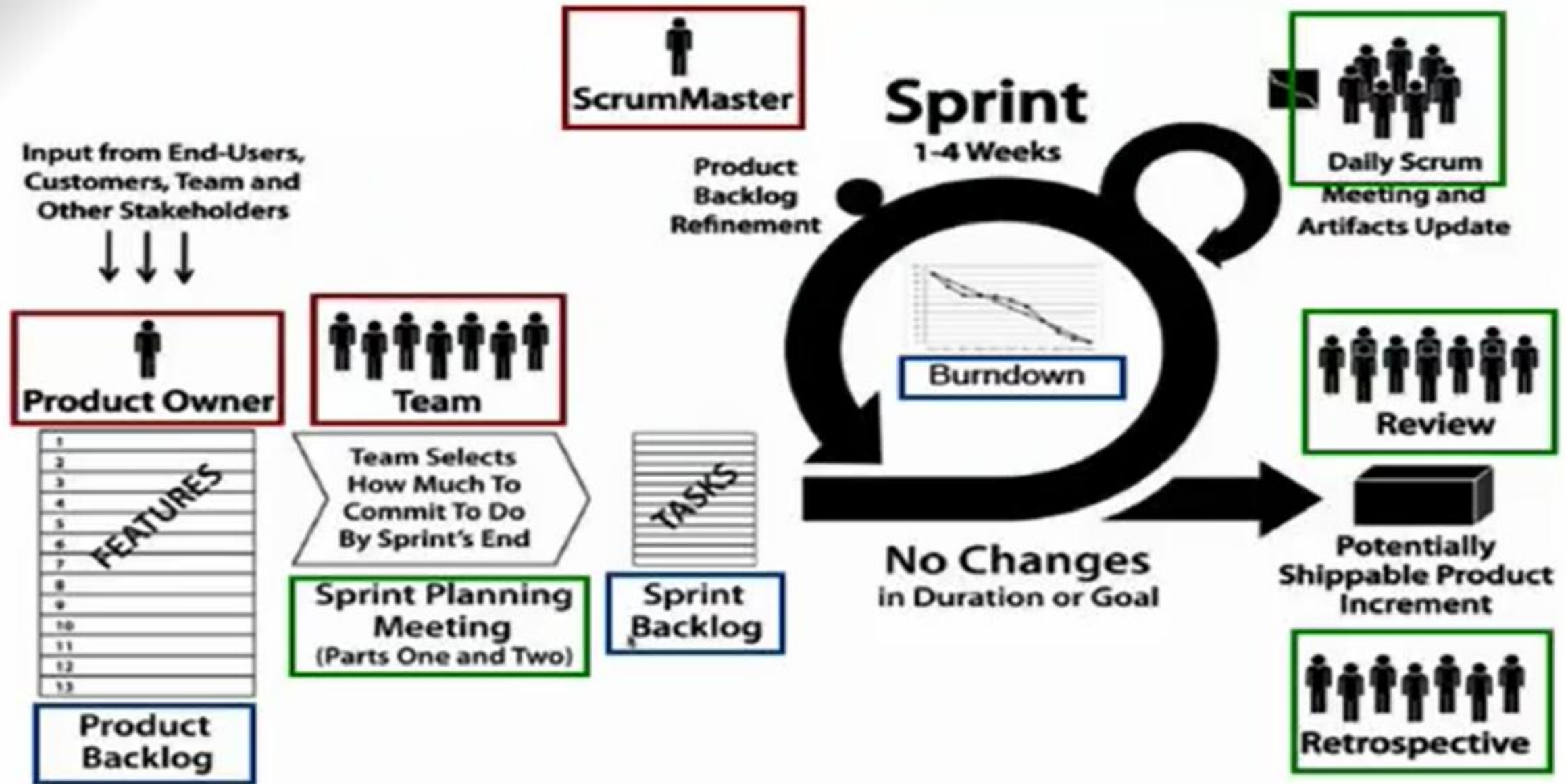
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SCRUM process in detail

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- Scrum principles are consistent with the agile manifesto and are used to guide development activities within a process that incorporates the following framework activities: requirements, analysis, design, evolution, and delivery.
- Within each framework activity, work tasks take place in a relatively short time-boxed period called a sprint.
- The work conducted within a sprint (the number of sprints required for each framework activity will vary depending on size of the product and its complexity) is adapted to the problem at hand and is defined and often modified in real time by the Scrum team

- It is **an iterative, incremental** framework
- Sprints – cycles of work developed, duration: 2 - 4 weeks; occur one after another **without pause**
- **Time boxed** – they end whether or not the work ends
- At the beginning, cross-functional team forms the priority list based on **customer requirements**
- During the sprint the chosen items **do not change**
- Everyday inspection and adjustment
- End of the sprint, review with stakeholders
- **Feedbacks** are taken and incorporated into the sprint
- End of sprint, **fully tested product** is formed as per customer requirements



THANK YOU

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